

YuvaSaarthi 2.0 - Presentation Cheatsheet

YuvaSaarthi 2.0 - Presentation Cheatsheet & Documentation

1. Project Overview

Title: YuvaSaarthi 2.0 - National AI-Powered Multilingual Education Assistant

Mission: To democratize access to educational guidance for every Indian student, bridging the language and information gap.

Tagline: "Empowering Indian Students with Knowledge."

2. The Problem Statement

- **Language Barrier:** Most quality educational content is in English. Rural students struggle to access it.
- **Information Overload:** Students are confused by dispersed information about exams, scholarships, and careers.
- **Lack of Personal Guidance:** 1 counselor per 100,000 students in India.

3. The Solution: YuvaSaarthi

- A **Multilingual AI Chatbot** that speaks 23 Indian Languages.
- Acts as a personal counselor for:
 - **Exams:** JEE, NEET, UPSC, Boards.
 - **Admissions:** College processes, eligibility.
 - **Concept Learning:** Explains complex topics simply.
 - **Scholarships:** Finds financial aid.

4. Key Features (The "Wow" Factors)

1. Bharat-First Language Support:

- Supports **23 Official Languages** (Hindi, Kannada, Tamil, Marathi, Rajasthani, etc.).
- Uses specialized "Bhashini" concepts (Government of India initiative emulation).
- **Dynamic Keyboard:** Telegram bot allows instant language switching.

2. Hybrid RAG Architecture (Retrieval Augmented Generation):

- Doesn't just hallucinate; retrieves facts from a curated knowledge base (PDFs, Docs).

YuvaSaarthi 2.0 - Presentation Cheatsheet

- If answer is not in docs, uses LLM intelligence (Llama-3-70b).

3. Video Integration:

- "Don't just read, watch."
- Automatically finds highly relevant **YouTube Tutorials** for concepts.
- Filters for educational quality (excludes shorts).

4. Omni-Channel Availability:

- **Web App:** Modern, Glassmorphism UI (Next.js).
- **Telegram Bot:** Accessible on low-end devices without installing new apps.

5. Technology Stack (Technical Deep Dive)

- **Frontend:**
 - Next.js 16 (React Framework)
 - Tailwind CSS (Styling)
 - Lucide React (Icons)
 - Glassmorphism Design System
- **Backend (The "Brain"):**
 - **FastAPI:** High-performance Python web server.
 - **LangChain:** Orchestrates the AI logic.
 - **ChromaDB:** Vector Database to store and retrieve knowledge.
- **AI & ML Models:**
 - **LLM:** Llama-3.3-70b-versatile (via Groq API) for reasoning.
 - **Embeddings:** `paraphrase-multilingual-mpnet-base-v2` (Best for Indian languages).
 - **Translation:** `deep-translator` (Google Translate API).
- **Integrations:**
 - **Telegram Bot API:** For the chat interface.
 - **YouTube Data API:** For video recommendations.

YuvaSaarthi 2.0 - Presentation Cheatsheet

6. System Architecture Flow

1. User asks Query (e.g., "JEE dates?" in Hindi).
2. Language Detection: System identifies "Hindi".
3. Translation: Translates query to English ("What are JEE dates?").
4. Vector Search: ChromaDB finds the official NTA notification doc.
5. RAG Generation: LLM generates answer using the doc.
6. Translation: Answer translated back to Hindi.
7. Video Search: Fetches "JEE Preparation" videos.
8. Response: User gets Hindi text + Video Links.

7. Setup & Deployment (How to Run)

1. Prerequisites: Python 3.9+, Node.js 18+.
2. Environment Variables: ``.env`` file with Keys (Groq, Telegram, YouTube).
3. Backend: ``python3 api_server.py`` (Runs on Port 8000).
4. Frontend: ``npm run dev`` (Runs on Port 3000).
5. Bot: ``python3 telegram_bot.py``.

8. Future Roadmap

- **Voice Mode:** Speak to the bot in native dialect.
- **Image Analysis:** scan homework questions to get answers.
- **Personalized Learning Paths:** AI generates a 3-month study plan.

9. Conclusion

YuvaSaarthi is not just a chatbot; it's a Digital Public Good for India's education sector, ensuring no student is left behind due to language or geography.